

Fungi and single-celled life

LIFE ON EARTH INCLUDES ORGANISMS THAT ARE NEITHER ANIMAL NOR PLANT.

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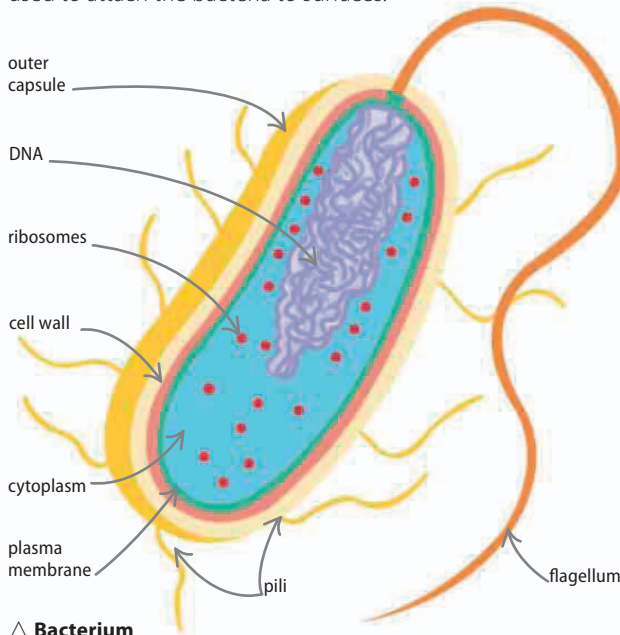
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The life forms within the Bacteria and Archaea domains, and most of the protist kingdom, are single-celled and can be viewed only through a microscope. By contrast, members of the fungi kingdom can grow into the largest organisms in the natural world.

Bacteria

The cells of Bacteria are hundreds of times smaller than those of plants or animals. They do not have a nucleus. Instead, their DNA is stored as a tangled loop called a plasmid. There are no other large organelles bound by a membrane, and all the metabolic reactions occur in the cytoplasm. Many bacteria move by flapping a whiplike flagellum. The hairlike pili are used to attach the bacteria to surfaces.



△ Bacterium

Most bacteria are surrounded by three layers. The plasma membrane is similar to the one in other types of cell. The cell wall is made of proteins and sugars. The starchy outer capsule, which stops the cell from drying out, is missing in some species.

A **honey fungus** in Oregon, USA, is nearly 9 sq km (3.5 sq miles) in area, making it the **largest** single organism on Earth.

Archaea

For many years, these microorganisms were considered to be types of Bacteria, and the two groups were classified together. However, recent DNA analysis suggests that Archaea are a totally separate group. Many archaea are extremophiles—they survive in extreme conditions, such as incredibly hot or cold places. It is likely that their ancestors evolved in the extreme habitats of the young Earth about 3.5 billion years ago.

▷ Haloquadratum

This archaea lives in brine pools, where the salt content kills most other life forms. It has a square cell (its name means “salt square”) filled with gas bubbles that help it float. No one knows how the cell survives.



▽ Pyrococcus

Discovered in the super-hot water that gushes from hydrothermal vents on the deep ocean floor, this archaea's name means “fire sphere.” Sunlight never reaches its habitat, and the archaea is sustained by chemicals in the hot water.

